

2007 级概率论与数理统计试卷 A 卷参考答案

一、

1. C

注释：由“ $A \subset B$ 成立”得 $P(A) = P(AB)$

$$\text{故 } P(A|B) = \frac{P(AB)}{P(B)} = \frac{P(A)}{P(B)}$$

2. C 3. B 注释：参考课本 86 页 4. B 注释： $\int_0^{2\pi} 2 \sin x dx = 1$?5.

6. B A 项参见课本 64 页，D 项参见课本 86 页

二、1. 2 注释：若 X 服从 Poisson 分布，则 $EX = \lambda$ ， $DX = \lambda$ 。（课本 84 页）

2. 12 注释： $\text{cov}(X, Y) = r_{XY} \cdot \sqrt{DX \cdot DY}$ 。（参考课本 86 页）

3. 1/5 注释：运用等比求和公式 $S = \frac{a_1(1-q^n)}{1-q}$

4. 38.4 注释： $E(\xi^2) = D\xi + (E\xi)^2$ ，对于 $\xi \sim B(n, p)$ ， $E\xi = np$ ， $D\xi = npq$

$$5. p(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0 \\ 0, & x \leq 0 \end{cases}, E\xi = \frac{1}{\lambda}, D\xi = \frac{1}{\lambda^2}$$

6. 0.2 注释：类似 2006 级试卷填空题第 6 题 7. 2/5

三、(1) 1/20; (2) 14/15

注释：(1) $P(A) = \frac{C_4^2}{C_{10}^3}$ ， C_4^2 表示从 7、8、9、10 这四个数中选两个；

(2) $\overline{B}$ = “三个号码中既含 4 又含 6”

四、(1) C=4; (2) $P\{\xi + \eta < 1\} = \int_0^1 dx \int_0^{1-x} 4e^{-2(x+y)} dy = 1 - 3e^{-2}$;

(3)

$$p_\xi(x) = \begin{cases} 2e^{-2x} & x \geq 0 \\ 0 & x < 0 \end{cases}, p_\eta(y) = \begin{cases} 2e^{-2y} & y \geq 0 \\ 0 & y < 0 \end{cases}$$

因 $p_\xi(x) \cdot p_\eta(y) = p(x, y)$ ，故 ξ 与 η 独立

? (4)

ξ 与 η 独立，所以 $\text{cov}(\xi, \eta) = 0$

$$E\xi = \int_0^{+\infty} x \cdot 2e^{-2x} dx = \frac{1}{2}, E\xi^2 = \int_0^{+\infty} x^2 \cdot 2e^{-2x} dx = \frac{1}{2}$$

$$\text{故 } D\xi = E\xi^2 - (E\xi)^2 = \frac{1}{4}$$

$$\text{同理, } E\eta = \frac{1}{2}, D\eta = \frac{1}{4}$$

五、 0.9979 注释：运用全概率公式，类似 2006 级试卷第三题

六、 0.9525

注释：设这100个部件中没有损坏部件数为 X ，

则 X 服从二项分布 $B(100, 0.9)$ ，且有

_____ $EX=np=100 \times 0.9=90$, $DX=npq=90 \times 0.1=9$

由拉普拉斯定理，

$$P\{a < X < b\} \approx \Phi\left(\frac{b-EX}{\sqrt{DX}}\right) - \Phi\left(\frac{a-EX}{\sqrt{DX}}\right)$$

故至少须有85个部件工作的概率为：

$$P\{X \geq 85\} \approx 1 - \Phi\left(\frac{85-90}{\sqrt{9}}\right) = 1 - \Phi(-1.67) = \Phi(1.67) = 0.9525$$

七、 $M=160$

注释：设出事人数为 X ，则有 $X \sim B(5000000, 0.0003)$

$EX=5000000 \times 0.0003=1500$, $DX=5000000 \times 0.0003 \times 0.9997 \approx 1500$

若要以99%的概率保证保险公司在此项保险中获得60万元以上的利润，

则 $P\{5000000M \times (1-40\%) - X \times 300000 \geq 600000\} \geq 99\%$

得 $P\{X \leq 10M-2\} \geq 99\%$

故需满足 $P\left\{\frac{X-1500}{\sqrt{1500}} \leq \frac{10M-2-1500}{\sqrt{1500}}\right\} \geq 99\%$

即 $\Phi\left(\frac{10M-2-1500}{\sqrt{1500}}\right) \geq 99\% \approx \Phi(2.33)$

解得 $M \geq 159.22$ ，故 $M = 160$

八、 (1) 课本 98 页辛欣大数定理

$$\text{由于 } E(\xi_n) = 0 \cdot \left(1 - \frac{2}{n}\right) + \sqrt{n} \cdot \frac{1}{n} + (-\sqrt{n}) \cdot \frac{1}{n} = 0$$

$$D(\xi_n) = E(\xi_n^2) - [E(\xi_n)]^2 = E(\xi_n^2)$$

$$\text{_____} = 0 \cdot \left(1 - \frac{2}{n}\right) + (\sqrt{n})^2 \cdot \frac{1}{n} + (-\sqrt{n})^2 \cdot \frac{1}{n} = 2$$

$$\text{令 } \bar{\xi}_n = \frac{1}{n} \sum_{k=2}^{n+1} \xi_k, k=2, 3, \dots, \text{ 则 } E(\bar{\xi}_n) = \frac{1}{n} \sum_{k=2}^{n+1} E(\xi_k) = 0$$

$$(2) \text{ _____ } D(\bar{\xi}_n) = \frac{1}{n^2} \sum_{k=2}^{n+1} D(\xi_k) = \frac{1}{n^2} \cdot 2n = \frac{2}{n}$$

由契比雪夫不等式，对任意的 $\varepsilon > 0$ ，有

$$\text{_____} P\{|\bar{\xi}_n - E(\bar{\xi}_n)| < \varepsilon\} \geq 1 - \frac{2}{n\varepsilon^2}$$

故有 $\lim_{n \rightarrow \infty} P\{|\bar{\xi}_n - E(\bar{\xi}_n)| < \varepsilon\} = 1$

即 $\{\xi_n\}$ 服从大数定律

2008 年概率论与数理统计试卷 A 卷参考答案

一、1.D 注释: $P(X \leq 1+u) = P(\frac{X-u}{\sigma} \leq \frac{1+u-u}{\sigma}) = \Phi(\frac{1}{\sigma})$

2.C 注释: 参考课本第 8 页

3.A 注释: 连续型随机变量在某一个点上的概率取值为零, 故 A 正确

? B 项是否正确

4.B 注释: 参考课本 86 页 5.A

二、 1. 1.33(或者填 $\frac{1359}{1024}$) 2. 25 注释: 参考课本 86 页 3. 0.25

4. $(X+Y) \sim B(7, p)$

注释: $E(X)=3p, E(Y)=4p$, 故 $E(X+Y)=E(X)+E(Y)=3p+4p=7p$;

$D(X)=3p(1-p), D(Y)=4p(1-p)$ 且 X, Y 独立, 故 $D(X+Y)=D(X)+D(Y)=3p(1-p)+4p(1-p)$

$$\text{设 } (X+Y) \sim B(n, P), \text{ 则有 } \begin{cases} E(X+Y)=7p=nP \\ D(X+Y)=3p(1-p)+4p(1-p)=nP(1-P) \end{cases}$$

解得 $n=7, P=p$

5. 2/5

X 的密度函数为 $f(x) = \frac{1}{5}$

方程有实根, 则必须满足 $\Delta = b^2 - 4ac = (2X)^2 - 4 \times 1 \times (5X - 4) \geq 0$,

即 $X \leq 1$ 或者 $X \geq 4$.

故方程有实根的概率 $P = \int_0^1 \frac{1}{5} dx + \int_4^5 \frac{1}{5} dx = \frac{2}{5}$

6. 0.35

由 $E(3X+5)=11$ 得 $EX=2$

由 $D(3X+5)=\sigma^2$ 得 $DX = \frac{\sigma^2}{9}, \sqrt{DX} = \frac{\sigma}{3}$

因 $P\{2 < X < 4\} = 0.15$, 故 $P\{\frac{2-2}{\frac{\sigma}{3}} < \frac{X-2}{\frac{\sigma}{3}} < \frac{4-2}{\frac{\sigma}{3}}\} = 0.15$

所以 $\Phi(\frac{2}{\frac{\sigma}{3}}) - \Phi(0) = 0.15$

所以 $\Phi(\frac{2}{\frac{\sigma}{3}}) - \Phi(\frac{-2}{\frac{\sigma}{3}}) = 0.3$

$P\{X < 0\} = P\{\frac{X-2}{\frac{\sigma}{3}} < \frac{0-2}{\frac{\sigma}{3}}\} = \Phi(\frac{-2}{\frac{\sigma}{3}}) = [1 - \Phi(\frac{2}{\frac{\sigma}{3}}) - \Phi(\frac{-2}{\frac{\sigma}{3}})]/2$

_____ = $[1 - 0.3]/2 = 0.35$

27. 相关

三、

设A=“取出的产品是正品”；

$B_{\text{甲}}$ =“取出的产品是甲厂生产的”

$B_{\text{乙}}$ =“取出的产品是乙厂生产的”

$B_{\text{丙}}$ =“取出的产品是丙厂生产的”

则 $P(A) = P(AB_{\text{甲}}) + P(AB_{\text{乙}}) + P(AB_{\text{丙}})$

$$= 0.5 \times 0.9 + 0.3 \times 0.8 + 0.2 \times 0.7 = 0.83$$

$$P(B_{\text{甲}} | A) = \frac{P(AB_{\text{甲}})}{P(A)} = \frac{P(B_{\text{甲}}) \cdot P(A | B_{\text{甲}})}{P(A)} = \frac{0.5 \times 0.9}{0.83} = 0.54$$

四、

$$X \sim \begin{pmatrix} -1 & 1 & 3 \\ 0.3 & 0.5 & 0.2 \end{pmatrix}$$

$$EX = (-1) \times 0.3 + 1 \times 0.5 + 3 \times 0.2 = 0.8$$

五、

$$(1) \text{由题意} f(x) = \begin{cases} \frac{1}{2}e^x & x \leq 0 \\ \frac{1}{2}e^{-x} & x > 0 \end{cases}$$

$$\text{故} F(x) = \begin{cases} \frac{1}{2}e^x & x \leq 0 \\ 1 - \frac{1}{2}e^{-x} & x > 0 \end{cases}$$

$$(2) P(-5 < X < 10) = (1 - \frac{1}{2}e^{-10}) - (\frac{1}{2}e^{-5})$$

$$(3) EX = \int_{-\infty}^0 x \cdot \frac{1}{2}e^x dx + \int_0^{+\infty} x \cdot \frac{1}{2}e^{-x} dx = \frac{1}{2}[(x-1)e^x]_{-\infty}^0 + \frac{1}{2}[(-x-1)e^{-x}]_0^{+\infty} \\ = 0$$

$$EX^2 = \int_{-\infty}^0 x^2 \cdot \frac{1}{2}e^x dx + \int_0^{+\infty} x^2 \cdot \frac{1}{2}e^{-x} dx$$

$$= \frac{1}{2}[x^2 e^x - 2xe^x + 2e^x]_{-\infty}^0 + \frac{1}{2}[-x^2 e^{-x} - 2xe^{-x} - 2e^{-x}]_0^{+\infty} = 2$$

$$DX = EX^2 - (EX)^2 = 2$$

设 X_i 为第 i 台机床生产的次品率

$$X_i \sim U(0.005, 0.035)$$

$$EX_i = \frac{0.005 + 0.035}{2} = 0.02, DX_i = \frac{1}{12}(0.035 - 0.005)^2 = 0.000075$$

? 六、(注: 对于均匀分布 $X_i \sim U(a, b)$, 有 $EX_i = \frac{a+b}{2}$, $DX_i = \frac{1}{12}(b-a)^2$)

$$\text{设总次品率 } Y = \sum_{i=1}^{2000} X_i$$

若要满足这批产品的平均次品率小于 0.025, 则 $Y < 0.025 \times 2000 = 50$

$$P\{Y < 50\} = P\left\{\frac{Y - 2000 \times 0.02}{\sqrt{2000 \times 0.000075}} < \frac{50 - 2000 \times 0.02}{\sqrt{2000 \times 0.000075}}\right\} = \Phi(25.8)$$

? 试卷中没有给出 $\Phi(25.8)$ 的值, 且直观上感觉 $\Phi(25.8)$ 的值太大了, 故不能肯定题中的做法是否可行

七、

$$(1) S_{\text{椭圆}} = \pi ab$$

$$\text{故 } (x, y) \text{ 的联合密度函数 } f(x, y) = \begin{cases} \frac{1}{\pi ab} & -a \leq x \leq a, -b \leq y \leq b \\ 0 & \text{其它} \end{cases}$$

$$X \text{ 的边缘密度函数 } f_X(x) = \begin{cases} \frac{2}{\pi a} & -a \leq x \leq a \\ 0 & \text{其它} \end{cases}$$

$$Y \text{ 的边缘密度函数 } f_Y(y) = \begin{cases} \frac{2}{\pi b} & -b \leq y \leq b \\ 0 & \text{其它} \end{cases}$$

$$(2) EX = \int_{-a}^a x \cdot \frac{2}{\pi a} dx = 0, EY = \int_{-b}^b y \cdot \frac{2}{\pi b} dy = 0,$$

$$EX^2 = \int_{-a}^a x^2 \cdot \frac{2}{\pi a} dx = \frac{4a^2}{3\pi}, EY^2 = \int_{-b}^b y^2 \cdot \frac{2}{\pi b} dy = \frac{4b^2}{3\pi}$$

$$DX = EX^2 - (EX)^2 = \frac{4a^2}{3\pi} = 25, DY = EY^2 - (EY)^2 = \frac{4b^2}{3\pi} = 4$$

$$\text{解得 } a = \frac{5\sqrt{3\pi}}{2}, b = \sqrt{3\pi}$$

$$(3) -a \leq x \leq a, -b \leq y \leq b \text{ 时,}$$

$$f_X(x) \cdot f_Y(y) = \frac{2}{\pi a} \cdot \frac{2}{\pi b} \neq \frac{1}{\pi ab}, \text{ 故 } X \text{ 与 } Y \text{ 不独立}$$

八、

Z 的分布函数 $F(z) = P\{Z \leq z\} = 1 - P\{Z > z\} = 1 - P\{\min(X, Y) > z\}$

$$= 1 - P(X > z, Y > z) = 1 - P(X > z) \cdot P(Y > z)$$

当 $z \leq 0$ 时, $P(X > z) = P(Y > z) = 1$

$$\text{故 } F(z) = 1 - 1 = 0$$

$$\text{当 } 0 < z \leq 1 \text{ 时, } P(X > z) = \int_z^1 1 dx = 1 - z$$

$$P(Y > z) = \int_z^1 5e^{-5x} dx = e^{-5z} - e^{-5}$$

$$\text{故 } F(z) = 1 - (1 - z) \cdot (e^{-5z} - e^{-5})$$

当 $z > 1$ 时, $P(X > z) = 0$

$$\text{故 } F(z) = 1 - 0 = 1$$

$$\text{所以 } F(z) = \begin{cases} 0 & z \leq 0 \\ 1 - (1 - z) \cdot (e^{-5z} - e^{-5}) & 0 < z \leq 1 \\ 1 & z > 1 \end{cases}$$

$$f(z) = \begin{cases} 0 & z \leq 0 \\ 6e^{-5z} - 5ze^{-5z} - e^{-5} & 0 < z \leq 1 \\ 0 & z > 1 \end{cases}$$

2009 年 2 学分参考答案

一、解：设 $A_i = \{\text{第 } i \text{ 枚弹道导弹击沉航空母舰}\}$, $B_i = \{\text{第 } i \text{ 枚弹道导弹击伤航空母舰}\}$

$C_i = \{\text{第 } i \text{ 枚弹道导弹没有击中航空母舰}\}$, $i=1, 2, 3, 4$

$D = \{\text{发射 4 枚弹道导弹能击沉航空母舰}\}$

$$P(A_i) = \frac{1}{3}, \quad P(B_i) = \frac{1}{2}, \quad P(C_i) = \frac{1}{6}, \quad i=1, 2, 3, 4$$

$$\bar{D} = C_1 C_2 C_3 C_4 \cup B_1 C_2 C_3 C_4 \cup C_1 B_2 C_3 C_4 \cup C_1 C_2 B_3 C_4 \cup C_1 C_2 C_3 B_4$$

$$P(\bar{D}) = P(C_1 C_2 C_3 C_4) + P(B_1 C_2 C_3 C_4) + P(C_1 B_2 C_3 C_4) + P(C_1 C_2 B_3 C_4) + P(C_1 C_2 C_3 B_4)$$

$$= \left(\frac{1}{6}\right)^4 + 4 \times \left(\frac{1}{6}\right)^3 \times \frac{1}{2} = \frac{13}{6^4}$$

$$P(D) = 1 - P(\bar{D}) = 1 - \frac{13}{6^4} = 0.99$$

二、解：(1) $A = \{\text{同花顺 (5 张同一花色连续数字构成)}\}$

$$P(A) = \frac{4 \times (13 - 4)}{C_{52}^5} = \frac{36}{C_{52}^5} \quad (\text{只要说明顺子的构成, 分子 40 也算对})$$

(2) $A = \{\text{3 张带一对 (3 张数字相同、2 张数字相同构成)}\}$

$$P(A) = \frac{C_{13}^1 C_4^3 C_{12}^1 C_4^2}{C_{52}^5}$$

(3) A = {3 张带 2 散牌 (3 张数字相同、2 张数字不同构成)}

$$P(A) = \frac{C_{13}^1 C_4^3 C_{12}^2 C_4^1 C_4^1}{C_{52}^5}$$

三、解：(1) 设 A = { 被检查后认为是非危险人物 }, B = { 过关的人是危险人物 }, 则

$$P(A) = P(B)P(A|B) + P(\bar{B})P(A|\bar{B}) = 0.96 \times 0.98 + 0.04 \times 0.05 = 0.9428$$

$$P(B|A) = \frac{P(B)P(A|B)}{P(A)} = 0.998$$

(2) 设需要 n 道卡, 每道检查系统是相互独立的, 则

$C_i = \{\text{第 } i \text{ 关危险人物被误认为非危险人物}\}$, $P\{C_1 \cdots C_n\} = 0.05^n$, 所以

$$1 - 0.05^n \geq 0.999, \quad n \geq \frac{\ln 0.0001}{\ln 0.05}, \quad \text{即 } n = \left\lceil \frac{\ln 0.0001}{\ln 0.05} \right\rceil + 1 = [3.0745] + 1 = 4$$

四、解：当 $a = 1$ 时, $Y = 1$, 则 $F_Y(y) = \begin{cases} 0 & y \leq 1 \\ 1 & y > 1 \end{cases}$

$$\text{当 } 0 < a < 1 \text{ 时, 当 } y \leq 0 \text{ 时, } F_Y(y) = P(Y < y) = 0, \quad f_Y(y) = \frac{dF_Y(y)}{dy} = 0$$

$$\text{当 } y > 0 \text{ 时, } F_Y(y) = P(a^X < y) = P(X \ln a < \ln y)$$

$$F_Y(y) = P\left(X > \frac{\ln y}{\ln a}\right) = 1 - P\left(X \leq \frac{\ln y}{\ln a}\right) = 1 - \Phi\left(\frac{\ln y}{\ln a}\right)$$

$$f_Y(y) = \frac{dF_Y(y)}{dy} = -\frac{1}{y \ln a} \cdot \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(\frac{\ln y}{\ln a} - \mu)^2}{2\sigma^2}}$$

$$\text{当 } a > 1 \text{ 时, 当 } y \leq 0 \text{ 时, } F_Y(y) = P(Y < y) = 0, \quad f_Y(y) = \frac{dF_Y(y)}{dy} = 0$$

$$\text{当 } y > 0 \text{ 时, } F_Y(y) = P\left(X < \frac{\ln y}{\ln a}\right) = \Phi\left(\frac{\ln y}{\ln a}\right)$$

$$f_Y(y) = \frac{dF_Y(y)}{dy} = \frac{1}{y \ln a} \cdot \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(\frac{\ln y}{\ln a} - \mu)^2}{2\sigma^2}}$$

五、解：(1) $E(X+Y) =$

$$= -3a - 2 \times 0.14 - b - 1 \times 0.01 + 1 \times 0.03 + 1 \times 0.13 + 2 \times 0.14 + 3 \times 0.15$$

$$= -3a - b + 0.6 = 0$$

$$a + 0.14 + b + 0.01 + 0.02 + 0.03 + 0.12 + 0.13 + 0.14 + 0.15 = a + b + 0.74 = 1$$

联立解得： $a = 0.17$, $b = 0.09$

(2) X 的概率分布函数:

X	-2	-1	0	1
	0.17	0.23	0.06	0.54

$$(3) E(XY) = 2 \times 0.17 + 1 \times 0.14 - 1 \times 0.12 + 1 \times 0.14 + 2 \times 0.15 = 0.8$$

$$\text{六、解: } P\left\{\left|\frac{m}{n} - p\right| < 0.1\right\} \geq 0.95, \text{ 因 } \frac{\frac{m}{n} - p}{\sqrt{\frac{p(1-p)}{n}}} \sim N(0,1)$$

$$P\left\{\frac{\left|\frac{m}{n} - p\right|}{\sqrt{\frac{p(1-p)}{n}}} < \frac{0.1}{\sqrt{\frac{p(1-p)}{n}}}\right\} \geq 0.95, \quad \frac{0.1}{\sqrt{\frac{p(1-p)}{n}}} \geq u_{0.975} = 1.96$$

$$n \geq (19.6)^2 p(1-p); \text{ 因为 } p(1-p) \leq 1/4, \text{ 取 } n \geq (19.6)^2 / 4 = 96.04 \text{ 即 } n = 97$$

七、解: (1) 二维随机变量(X,Y)的联合概率密度:

$$f(x, y) = \begin{cases} 1/ab, & 0 < x < a, 0 < y < b \\ 0, & \text{others} \end{cases}$$

$$\text{边缘概率密度: } f_X(x) = \begin{cases} 1/a, & 0 < x < a \\ 0, & \text{others} \end{cases}, \quad f_Y(y) = \begin{cases} 1/b, & 0 < y < b \\ 0, & \text{others} \end{cases}$$

$$(2) DX = (1/12)a^2 = 12, DY = (1/12)b^2 = 36, \quad a = 12, b = 12\sqrt{3}$$

(3) 随机变量 X 与 Y 相互独立, 因为 $f(x, y) = f_X(x)f_Y(y)$

$$\text{八、解: } P(|\xi| > t) = \int_{|x|>t} dF(x) \leq \int_{|x|>t} \frac{|x|^3}{t^3} dF(x) \leq \int_{|x|\geq 0} \frac{|x|^3}{t^3} dF(x) = \frac{E|\xi|^3}{t^3} = \frac{c}{t^3}$$

$$\text{九、解: (1) } \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x, y) dx dy = \int_0^1 \left(\int_0^1 Axy dy \right) dx = \frac{A}{4} = 1, \quad A = 4$$

$$(2) P(X < 0.4, Y < 1.3) = \int_0^{0.4} \left(\int_0^1 4xy dy \right) dx = 0.16$$

$$(3) Ee^{tX+sy} = \int_0^1 \left(\int_0^1 e^{tx+sy} 4xy dy \right) dx = \int_0^1 e^{tx} 4x \left(\left(\frac{ye^{sy}}{s} \right) \Big|_0^1 - \frac{1}{s} \int_0^1 e^{sy} dy \right) dx$$

$$= 4 \left(\frac{e^s}{s} - \frac{e^s}{s^2} + \frac{1}{s^2} \right) \left(\frac{e^t}{t} - \frac{e^t}{t^2} + \frac{1}{t^2} \right)$$

$$(4) \quad EX = \int_0^1 \left(\int_0^1 4x^2 y dy \right) dx = \frac{2}{3}, \quad EX^2 = \int_0^1 \left(\int_0^1 4x^3 y dy \right) dx = \frac{1}{2}$$

$$DX = EX^2 - (EX)^2 = \frac{1}{2} - \frac{4}{9} = \frac{1}{9}, \quad E(XY) = \int_0^1 \left(\int_0^1 4x^2 y^2 dy \right) dx = \frac{4}{9}$$

$$Cov(X, Y) = EXY - EX \cdot EY = \frac{4}{9} - \frac{2}{3} \times \frac{2}{3} = 0$$

十、解：(1) 设 ξ 表示该观众答对题数， $\xi = 0, 1, 2, \dots$

则第 $\xi+1$ 次解答答错（即首次出错）。

答对一题的概率为

$$\begin{aligned} P(\text{答对题}) &= P(\text{答对A题} | \text{选择A题})P(\text{选择A题}) + P(\text{答对B题} | \text{选择B题})P(\text{选择B题}) \\ &= 0.4 \times 0.5 + 0.6 \times 0.5 = 0.5 \end{aligned}$$

答错一题的概率为 0.5

$$\text{所以 } P(\xi = k) = 0.5^k \times 0.5 = 0.5^{k+1}; \quad E\xi = \sum_{k=0}^{\infty} k \cdot 0.5^{k+1} = 1$$

(2) 观众得到奖励金额 η 的期望值：

$$\text{令 } X = \begin{cases} 1, & \text{答对A题} \\ 2, & \text{答对B题} \\ 3, & \text{答错题} \end{cases}, \text{ 则 } X \sim \begin{pmatrix} 1 & 2 & 3 \\ 0.2 & 0.3 & 0.5 \end{pmatrix},$$

$$E\eta = E(E(\eta | X)) = 0.2 \times E(1000 + \eta) + 0.3 \times E(500 + \eta) + 0.5 \times 0$$

$$\therefore E\eta = 700$$

或：答对一题得到奖金的期望为： $0.5 \times 0.4 \times 1000 + 0.5 \times 0.6 \times 500 = 350$

进入第 k 题答题环节的概率为： 0.5^{k-1}

$$\text{因此，总奖金的期望为：} \sum_{k=1}^{\infty} 350 \times 0.5^{k-1} = 700$$

2010 年《概率论与数理统计》(2 学分)

$$\text{一. 解: } \frac{C_1^1 C_8^1}{C_9^2} = \frac{2}{9}$$

二. 解：设 $A = \{\text{从事某职业的可疑病人}\}$ ， $B = \{\text{患有肺癌}\}$

$$P(A) = 0.45, \quad P(\bar{A}) = 0.55, \quad P(B|A) = 0.9, \quad P(B|\bar{A}) = 0.05$$

$$(1) P(B) = P(A)P(B|A) + P(\bar{A})P(B|\bar{A}) = 0.4325$$

$$(2) P(A|B) = \frac{P(A)P(B|A)}{P(B)} = 0.9364$$

三. 解: 设 A = { 第一种工艺下的合格品 }, B = { 第二种工艺下的合格品 }, C = { 优质品 }

$$P(C|A) = 0.9, \quad P(C|B) = 0.8$$

D_i = { 表示第一种工艺下的第 i 道工序生产的废品 }, $i=1, 2, 3$

$$P(D_1) = 0.05, \quad P(D_2) = 0.10, \quad P(D_3) = 0.25$$

E_i = { 表示第二种工艺下的第 i 道工序生产的废品 }, $i=1, 2$

$$P(E_1) = P(E_2) = 0.10$$

$$P(A) = P(\bar{D}_1 \bar{D}_2 \bar{D}_3) = P(\bar{D}_1)P(\bar{D}_2)P(\bar{D}_3) = 0.95 * 0.90 * 0.75 = 0.6413$$

$$P(B) = P(\bar{E}_1 \bar{E}_2) = P(\bar{E}_1)P(\bar{E}_2) = 0.90 * 0.90 = 0.81$$

$$P(CA) = P(A)P(C|A) = 0.6413 * 0.9 = 0.5771$$

$$P(CB) = P(B)P(C|B) = 0.81 * 0.8 = 0.648$$

答: 第一种工艺得到的优等品概率大。

四. 解: 设 ξ 表示正常运行时间, $F(t) = P(\xi < t)$, 当 $t \leq 0$ 时, $F(t) = 0$

$t > 0$ 时, 题中条件为: $P(t < \xi < t + \Delta t | \xi > t) = \lambda \Delta t + o(\Delta t)$

$$\text{即: } \frac{P(t < \xi < t + \Delta t, \xi > t)}{P(\xi > t)} = \frac{F(t + \Delta t) - F(t)}{1 - F(t)} = \lambda \Delta t + o(\Delta t)$$

$$\frac{F(t + \Delta t) - F(t)}{\Delta t} = \lambda(1 - F(t)) + o(1)$$

$$\text{令 } \Delta t \rightarrow 0, \text{ 则 } \frac{dF(t)}{dt} = \lambda(1 - F(t))$$

$$\frac{d(1 - F(t))}{dt} = -\lambda(1 - F(t)), \quad 1 - F(t) = ce^{-\lambda t}, \quad \because F(0) = 0, \quad \therefore F(t) = 1 - e^{-\lambda t}$$

故它正常运行时间大于 t 概率: $P(\xi > t) = e^{-\lambda t}$

五. 解: ξ 的取值: $m, m+1, \dots$

$$(1) P(\xi = k) = C_{k-1}^{m-1} p^{m-1} (1-p)^{k-m} p = C_{k-1}^{m-1} p^m (1-p)^{k-m}$$

$$\xi = m, m+1, \dots$$

$$\begin{aligned} (2) \quad E(\xi) &= \sum_{k=m}^{\infty} k C_{k-1}^{m-1} p^m (1-p)^{k-m} \\ &= m \sum_{k=m}^{\infty} \frac{k!}{m!(k-m)!} p^m (1-p)^{k-m} = m \sum_{k=m}^{\infty} C_k^m p^m (1-p)^{k-m} \\ &= \frac{m}{p} \end{aligned}$$

或：令 η 为首次取到1位离婚人士的调查人数，则 η 服从几何分布，即

$$P(\eta = k) = (1-p)^{k-1} p, k=1, 2, \dots, \quad E\eta = \frac{1}{p}$$

设 $\eta_1, \eta_2, \dots, \eta_m$ 独立同分布于 η ，则 $\xi = \eta_1 + \dots + \eta_m$ ，得 $E\xi = \frac{m}{p}$

六. 解：(1) (ξ, η) 分布密度

$$f(x, y) = \begin{cases} \frac{1}{2}, & 1 \leq x \leq 2, 1 \leq y \leq 3 \\ 0, & \text{其它} \end{cases}$$

$$f_{\xi}(x) = \begin{cases} \int_1^3 \frac{1}{2} dy = 1, & 1 \leq x \leq 2 \\ 0, & \text{其它} \end{cases}$$

$$f_{\eta}(y) = \begin{cases} \int_1^2 \frac{1}{2} dx = 1/2, & 1 \leq y \leq 3 \\ 0, & \text{其它} \end{cases}$$

$$\begin{aligned} (2) \quad P(\xi < 1.5, \eta < 4) &= \int_1^{1.5} dx \int_1^3 \frac{1}{2} dy \\ &= \int_1^{1.5} dx = 0.5 \end{aligned}$$

$$(3) \quad \text{因 } f(x, y) = f_{\xi}(x) \cdot f_{\eta}(y)$$

所以，随机变量 ξ 与 η 是独立的。

七. 解：当 $y \leq 0$ 时， $F_{\eta}(y) = P\{\eta < y\} = P\{|\xi| < y\} = 0$,

所以, $y < 0$ 时, $f(y) = \frac{d}{dy} F_\eta(y) = 0$,

当 $y > 0$ 时, $F_\eta(y) = P\{\eta < y\} = P\{|\xi| < y\} = P(-y < \xi < y)$

$$= P(\xi < y) - P(\xi < -y) = P\left(\frac{\xi}{\sigma} < \frac{y}{\sigma}\right) - P\left(\frac{\xi}{\sigma} < -\frac{y}{\sigma}\right) = \Phi\left(\frac{y}{\sigma}\right) - \Phi\left(-\frac{y}{\sigma}\right)$$

$$f(y) = \frac{d}{dy} F_\eta(y) = \frac{1}{\sigma} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{\left(\frac{y}{\sigma}\right)^2}{2}} + \frac{1}{\sigma} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{\left(-\frac{y}{\sigma}\right)^2}{2}} = \frac{2}{\sigma\sqrt{2\pi}} e^{-\frac{y^2}{2\sigma^2}}$$

八. 解: 设把机器分解为 n 个部分, 各部分测量的误差分别为 $\xi_1, \xi_2, \dots, \xi_n$, 它们均应服从区间 $(-1, 1)$ 上的均匀分布。

$$E\xi_i = \int_{-1}^1 \frac{x}{2} dx = 0, \quad i = 1, 2, \dots, n, \quad E\xi_i^2 = \int_{-1}^1 \frac{x^2}{2} dx = \frac{x^3}{6} \Big|_{-1}^1 = \frac{1}{3}, \quad i = 1, 2, \dots, n$$

$$D\xi_i = E\xi_i^2 - (E\xi_i)^2 = \frac{1}{3}, \quad i = 1, 2, \dots, n$$

$$\text{即, 总误差: } \eta = \sum_{i=1}^n \xi_i, \quad E\eta = \sum_{i=1}^n E\xi_i = 0, \quad D\eta = \sum_{i=1}^n D\xi_i = \frac{n}{3}$$

由中心极限定理,

$$P\{|\eta| < 10\} = P\left\{\left|\frac{\eta}{\sqrt{\frac{n}{3}}}\right| < 10\sqrt{\frac{3}{n}}\right\} \approx \Phi\left(10\sqrt{\frac{3}{n}}\right) - \Phi\left(-10\sqrt{\frac{3}{n}}\right) \geq 0.99$$

$$10\sqrt{\frac{3}{n}} \geq u_{0.995} = 2.58, \quad n \leq 45.06$$

答: 最多可以把机器分解成 45 部分, 才能以不低于 99% 的概率保证测定的总重量误差的绝对值不超过 10kg。

九. 解: 由契比雪夫不等式, 则

$$P\left(\left|\frac{1}{n} \sum_{i=1}^n \xi_i - \frac{1}{n} \sum_{i=1}^n E(\xi_i)\right| > \varepsilon\right) \leq \frac{D\left(\frac{1}{n} \sum_{i=1}^n \xi_i\right)}{\varepsilon^2} = \frac{\frac{1}{n^2} \sum_{i=1}^n D\xi_i}{\varepsilon^2} \rightarrow 0$$

得

$$\lim_{n \rightarrow \infty} P\left(\left|\frac{1}{n} \sum_{i=1}^n \xi_i - \frac{1}{n} \sum_{i=1}^n E(\xi_i)\right| < \varepsilon\right) = 1$$